

EVANS KWASI

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EDUCATION

M.Sc. Computer Science | *University of Ghana, Accra*

Sep 2024 – Mar 2026

- Thesis: Towards the Development of a Maternal Health Chatbot: A Single-Speaker Model
- CGPA: 3.62 / 4.0 | Coursework: Machine Learning with Big Data Analytics, Advanced Computer Vision, Intelligent Systems, Distributed Systems, Information Systems Security, Bioinformatics

B.Sc. Computer Science – First Class | *KNUST, Kumasi*

Sep 2018 – Nov 2022

- CWA: 72.3 / 100 | Project: Deep Learning & IoT for Precision Agriculture — Automated Weed Detection and Multi-Sensor Soil Monitoring
- President, Computer Science Society

RESEARCH/PROFESSIONAL EXPERIENCE

Researcher / Developer | *Interactive Systems Research Group, York University*

Nov 2025 – Present

- Designing an adaptive mnemonic generation framework combining Multimodal LLMs and Genetic Algorithms to optimize memorability and long-term recall — directly applicable to personalised AI-driven decision support.
- Corresponding author: systematic review on persuasive technology in circular economy manuscript ready).
- Corresponding author: empirical study on cross-device interface consistency and user performance (in preparation).

Research Assistant & AI Engineer | *HCI Lab, University of Ghana*

Jun 2023 – Present

- Led development of Akan text-to-speech (VITS) and automatic speech recognition (Whisper) systems for low-resource Twi, in collaboration with Google Ghana — demonstrating user-adaptive NLP pipeline design.
- Co-led and developed an ASR system for people with speech impairment including Stammering, Cerebral Palsy, Cleft Palate.
- Co-developed a multilingual maternal health dataset (Twi, Fante, Ga, Ewe, Baoule) covering data collection, expert translation, and recording; manuscript in preparation for Data-in-Brief.
- Built Project Obaapayin: speech-to-speech maternal health chatbot with RAG pipeline, fine-tuned LLMs, and culturally grounded dialogue for low-resource settings — end-to-end AI system design.
- Designed and deployed lab website using React.js; built Twi spell checker using weighted Levenshtein distance.

Lead Data Scientist | *St. Michael's Hospital – Jackie Pramso*

Nov 2022 – May 2024

- Developed an explainable ML model for diabetes prediction (RF, SVM, XGBoost, ANN) trained on a custom Ghanaian clinical dataset; achieved 99.3% accuracy across classical models.
- Applied SHAP (SHapley Additive exPlanations) for both local and global model explainability—waterfall plots, mean SHAP feature importance, and dependence plots—enabling clinician-facing transparency.
- Conducted statistical validation using accuracy, precision, recall, F1-score, and ROC-AUC; contributed to academic publication of findings.

Privacy-Aware Synthetic Credit Risk Data Generation

Dec 2025

- Built a synthetic tabular data pipeline on the UCI Credit Card Default dataset, comparing Gaussian Copula, CTGAN, and TVAE for banking-style default prediction.
- Evaluated synthetic data using statistical fidelity, correlation preservation, TSTR predictive utility, nearest-neighbour privacy checks, and SHAP-based feature-importance comparison.

Backend Engineer (Part-Time) | *RxHealth Information Systems*

Sep 2021 – Jan 2023

- Built cross-platform beauty marketplace app (Flutter), integrated Paystack payment gateway, and designed an AI assistant for domain-specific health-related queries.

PUBLICATIONS & PRESENTATIONS

- Mensah, M. A., Kwasi, E., Abdulmawjood, M., & Oyibo, K. (2026, June). User Modelling and Adaptation in Immersive Preventive Health Interventions: A Systematic Review and Evidence Map. In *Proceedings of the 34th ACM Conference on User Modeling, Adaptation and Personalization* (pp. 263-272).

- Wiafe, I., Wiafe-Akenten, Kwasi, E., Ekpezu O.A., A. N., Appati, J. K., Odame, G., Yeboah, F., Mensah, M. A., Abdulai, J.-D., & Winful, F. B. P. (under review). Neural machine translation for maternal health communication in low-resource language: Akan-English. *Language Resources and Evaluation* [Special Issue: Less-Resourced Languages].
- Wiafe, I., Ekpezu, A. O., Winful, F. B. P., Wiafe-Akenten, A. N., Mensah, M. A., Semanyoh, N., Salihs, S. A., **Kwasi, E.**, Apatti, J. K., Odame, G., Osei, K. O., & Yeboah, F. E. (under review). A Parallel English–Akan Maternal Health Dataset to Support Machine Translation and Text-to-Speech Systems. *Data in Brief*, Elsevier
- Wiafe, I., **Kwasi, E.**, et al. (2025). Developing an Akan Text-to-Speech System: Corpus Creation, Neural Modeling, and Digital Inclusion in Ghana. Presented at SOLCON V, University of Ghana.
- Bekor, C., Wiafe, I., & Kwasi, E. (2025). AI-Powered Image Description in Akan and Ewe. Presented at SOLCON V, University of Ghana.
- Kwasi, E., et al. Detailed Report on Diabetes Explainability: Interpretable ML on a Ghanaian Clinical Dataset using SHAP. St. Michael's Hospital – Jackie Pramso.

TECHNICAL SKILLS

Python, JavaScript / React.js, Flutter, C++, MATLAB, SQL, PyTorch, TensorFlow / Keras, Scikit-learn

REFERENCES

Prof. Isaac Wiafe | University of Ghana | iwiafe@ug.edu.gh

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Dr. Mark Atta Mensah | University of Ghana | mamensah@ug.edu.gh